KYLE McCALGAN

FINAL-YEAR COMPUTER SCIENCE STUDENT

Pretoria, South Africa | 076 944 3986 | kylemccalgan@gmail.com
GitHub: https://github.com/KyleMcCalgan
LinkedIn: https://www.linkedin.com/in/kyle-mccalgan/



TECHNICAL SKILLS

Languages
Python, C++, Delphi, React, TypeScript; experience with Lua and GDScript

Web & Backend
React, Vite, Node.js, NestJS, FastAPI, REST APIs, WebSockets

Cloud
AWS architecture and deployment workflows

Data & Auth
PostgreSQL, Prisma, Supabase

DevOps & Tools
Docker, Git/GitHub, Linux/Ubuntu, WSL, Swagger/OpenAPI, MinIO

STRENGTHS

- Software design & modelling
- Problem solving & debugging
- Team-based development
- Technical documentation
- Computer troubleshooting

ACHIEVEMENTS

- Silver Certificate - 2022 Programming Olympiad
- Academic & Sporting Half-Colours
- Full scholarship to Durban High School
- First- and second-year distinction average at university

ADDITIONAL

Video editing
Premiere Pro, DaVinci Resolve, Wondershare Filmora

Interests
Hockey, golf, video editing

PROFILE

Final-year Computer Science student at the University of Pretoria with a strong academic record and practical experience building full-stack, containerised software in team environments. Comfortable across frontend, backend, databases, APIs, Linux and DevOps workflows, with a particular interest in software engineering, system design and security. Experienced in taking projects from architecture and implementation through testing, documentation and deployment planning.

FEATURED PROJECT

SpendSense - Personal Finance & Gamification Platform

COS301 Software Engineering Capstone | 2026 | 5-person team

github.com/COS301-SE-2026/SpendSense

- Developing a full-stack personal-finance platform combining obligations, payment tracking, simulated credit scoring, achievements, friends and challenge-based gamification.
- Working with React + Vite on the frontend, NestJS + Prisma on the backend, PostgreSQL, Supabase authentication, MinIO storage and a FastAPI AI service.
- Contributed to domain modelling, API design, Docker-based development, Swagger documentation, database migrations/seeding and automated GitHub CI workflows.
- Worked on deployment architecture using AWS services and a containerised application stack, with emphasis on maintainability and clear developer onboarding.

SELECTED PROJECTS

Multiplayer Kruger National Park boardgame recreation

Node.js • WebSockets

- Built a real-time multiplayer Kruger Park drive game using Node.js and WebSockets for synchronised game state and player interaction.

Interactive Nursery Simulation

C++ • raylib

- Created an interactive simulation-style game with a graphical interface using raylib.

Earlier projects: Wordle clone and object-oriented Snake game in Delphi.

EDUCATION

BSc Computer Science - University of Pretoria

Pretoria, South Africa | 2024–2026 (expected)

Academic performance: First- and second-year distinction average.

Significant modules: Imperative Programming (90%), Computer Organisation & Architecture (87%), Programme Design (86%), Data Structures & Algorithms (85%), Software Modelling (83%).

Durban High School

Durban, South Africa | 2019–2023

Bachelor's Pass with a distinction average. Selected subjects: Information Technology (92%), Geography (82%), Physical Science (80%).